Practice Set A

Class 12 Mathematics, Chapter 8: Application of Integrals

Maximum marks: **53** **20** questions

Name _____ Date _____ Score _____ / 53

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE SQP 2024-25 — 1 mark

A student observes an open-air honeybee nest on the branch of a tree, whose plane figure is parabolic in shape, given by $x^2 = 4y$. Then the area (in sq units) of the region bounded by the parabola $x^2 = 4y$ and the line $y = 4$ is:

(A) $\frac{32}{3}$ **(B)** $\frac{64}{3}$ **(C)** $\frac{128}{3}$ **(D)** $\frac{256}{3}$

2 MCQ, Board-pattern — 1 mark

The area of the region in the first quadrant enclosed by the circle $x^2 + y^2 = 36$ and the coordinate axes is:

(A) 9π sq units **(B)** 18π sq units

(C) 36π sq units **(D)** 6π sq units

3 MCQ, Board-pattern — 1 mark

The area enclosed by the ellipse $x^2 + 4y^2 = 16$ is:

(A) 16π sq units **(B)** 64π sq units

(C) 8π sq units **(D)** 4π sq units

4 MCQ, Board-pattern — 1 mark

The area of the region bounded by the curve $y = \sin x$, the x-axis and the lines $x = 0$ and $x = \pi$ is:

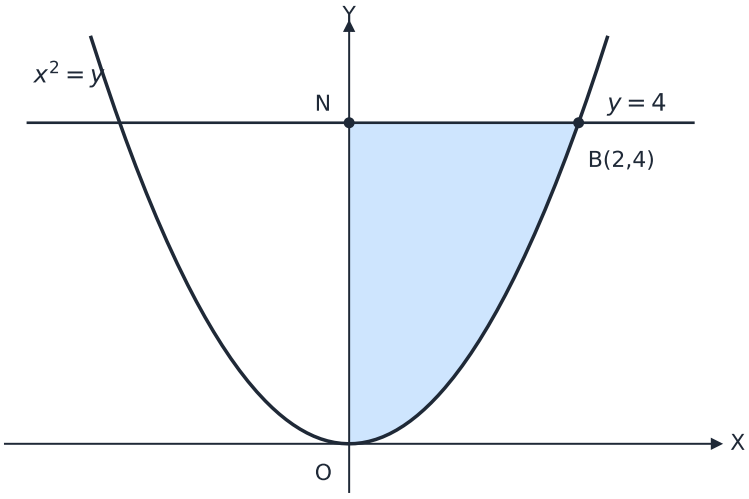
(A) 1 sq unit **(B)** 2 sq units **(C)** 0 **(D)** π sq units

5 MCQ, Board-pattern — 1 mark

The area of the region bounded by the line $x + 2y = 4$ and the coordinate axes is:

(A) 8 sq units **(B)** 2 sq units

(C) 16 sq units **(D)** 4 sq units

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** The area of the region bounded by the curve $y = x^3$, the x-axis and the lines $x = -1$ and $x = 1$ is 0. **Reason (R):** $\int_{-1}^{1} x^3\,dx = 0$, as x^3 is an odd function. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, CBSE SQP 2025-26 Find the area of the shaded region in the figure given below. The region lies in the first quadrant and is bounded by the parabola $x^2 = y$, the y-axis and the line $y = 4$. 	2 marks

Level 2

18 marks

8	Short answer, PYQ-style Sketch the graph of $y = x^2$ and, using integration, find the area of the region bounded by the curve $y = x^2$, the x-axis and the lines $x = -1$ and $x = 2$.	3 marks
9	Short answer, PYQ-style Sketch the region bounded by the lines $2x + y = 8$, $y = 2$, $y = 4$ and the y-axis. Hence, obtain its area using integration.	3 marks
10	Short answer, PYQ-style Using integration, find the area of the region bounded by the line $y = 4 - 2x$, the x-axis and the ordinates $x = -1$ and $x = 3$.	3 marks
11	Short answer, Board-pattern Using integration, find the area of the region bounded by the curve $y = \frac{3}{2}\sqrt{4 - x^2}$ and the x-axis.	2 marks

Q No.	Question	Marks
12	Short answer, PYQ-style Using integration, find the area of the minor segment of the circle $x^2 + y^2 = 16$ cut off by the line $x = 2$.	3 marks
13	Short answer, CBSE SQP 2024-25 Draw the rough sketch of the curve $y = 20\cos 2x$, where $\frac{\pi}{6} \le x \le \frac{\pi}{3}$. Using integration, find the area of the region bounded by the curve $y = 20\cos 2x$ from the ordinates $x = \frac{\pi}{6}$ to $x = \frac{\pi}{3}$ and the x-axis.	2 marks
14	Short answer, PYQ-style Using integration, find the area of the region bounded by the curve $y = x\|x\|$, the x-axis and the ordinates $x = -1$ and $x = 1$.	2 marks

Level 3 — 27 marks

15	Long answer, CBSE SQP 2026-27 Using integration, find the area of the region $\{(x, y) : 0 \le y \le \sqrt{4 - x^2},\ x \ge 2 - y\}$.	5 marks
16	Long answer, PYQ-style Using integration, find the area of the triangle whose vertices are $A(-1, 0)$, $B(1, 3)$ and $C(3, 2)$.	5 marks
17	Long answer, CBSE SQP 2026-27 Using integration, find the area of the region bounded by the curve $y^2 = 2x$ and the line $x - 2y = 0$.	5 marks
18	Long answer, CBSE SQP 2026-27 Using integration, find the area of the region enclosed within $25x^2 + 16y^2 = 400$.	5 marks
19	Short answer, CBSE SQP 2025-26 Using integration, find the area of the region $\{(x, y) : x^2 - 4y \le 0,\ y - x \le 0\}$.	3 marks
20	Case study, Board-pattern **Case Study:** A circular garden in a housing society has its boundary given by $x^2 + y^2 = 100$, with its centre at the origin (units in metres). A flower bed is to be made in the part of the garden between two straight paths along the lines $x = -5$ and $x = 5$. Based on this information, answer the following questions using integration.	4 marks

(i) Find the total area of the garden. — 1 mark

(ii) Find the area of the part of the garden lying in the first quadrant. — 1 mark

(iii) Find the area of the flower bed, that is, the part of the garden between the lines $x = -5$ and $x = 5$. — 2 marks

Answer key

Chapter 8: Application of Integrals, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $\frac{64}{3}$
2	(A) 9π sq units
3	(C) 8π sq units
4	(B) 2 sq units
5	(D) 4 sq units
6	(D) A is false, but R is true
7	$\frac{16}{3}$ sq units

Level 2

8	3 sq units
9	5 sq units
10	10 sq units
11	3π sq units
12	$\left(\frac{16\pi}{3} - 4\sqrt{3}\right)$ sq units
13	$(20 - 10\sqrt{3})$ sq units
14	$\frac{2}{3}$ sq units

Level 3

15	$(\pi - 2)$ sq units
16	4 sq units
17	$\frac{16}{3}$ sq units
18	20π sq units
19	$\frac{8}{3}$ sq units
20	(i) $100\pi\ \mathrm{m}^2$ (ii) $25\pi\ \mathrm{m}^2$ (iii) $\left(50\sqrt{3} + \frac{100\pi}{3}\right)\ \mathrm{m}^2$